

STEM

DEEP LEARNING

Neural Network Architecture & Backpropagation

Comprehensive structured revision guide with learning outcomes & analysis.

MODULE TYPE

AUDIENCE

WORD COUNT

TARGET EXAM

COMPREHENSIVE

EXPERT

3200 WORDS

GENERAL EXAM

1. 1. Computation Graph and Forward Pass

Deep neural networks execute sequential tensor operations across parameterized layers.



Hidden1 --> Activation[ReLU Activation] Activation --> Hidden2[Dense Layer 2] Hidden2
Loss]

1.1 Gradient Backpropagation

The chain rule decomposes compound matrix gradients: $dL/dW = dL/dy \cdot dy/dz \cdot dz/dW$

PYTHON

```
import torch
import torch.nn as nn

class Net(nn.Module):
    def __init__(self):
        super().__init__()
        self.fc1 = nn.Linear(784, 128)
        self.relu = nn.ReLU()
        self.fc2 = nn.Linear(128, 10)

    def forward(self, x):
        return self.fc2(self.relu(self.fc1(x)))
```

KEY

Gradient vanishing occurs when activation derivative saturation contracts backpropagating error vectors.